Simulation Study: Implementation

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Synthetic Data: Academics moving in US states

The following simulation study includes: - First part: REM including a linear effect for reciprocity and a non-linear effect for distance; - Second part: REM including a linear effect for reciprocity and a non-linear effect for distance plus a global effect for the day of the week.

Preliminarities

Members, Size, and Order of the temporal network

```
source("script/states.R")
```

```
##   |                                                                      |
```

Distance is an example of exogenous information that is external to the relational event network. We exploit this information to compute an endogenous covariate.

Source for distance: [Walker, 2024] [Pebesma, 2018] [Hijmans, 2022]

We assume the contribution of distance being a non-linear function of distance itself.

The true value for the coefficients is retrieved in such a way the final contribution is similar to a negative contribution of the logarithm of the original covariate, namely:

```
source("script/input_distance.R")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Linear effect of weekday and reciprocity

The baseline of a interstate academic conference is assumed to be smaller during the weekends with respect to the weekdays.

```
source("script/true_coefficients.R")
```

Event Simulation (no shifting)

```
event_simulator <- function(states, p, n, b0, b1, fit, seed) {
```

```r
set.seed(seed)
print(seed)

# some constants
current.tm<-0
rec<-matrix(0,ncol=p,nrow=p)

#where to save
dat<-NULL
nondat<-NULL
tms<-NULL
tm = 0  # Initialize time to 0

# Initialize the invaded list with a random sample of states
last_contacted <- matrix(NA, nrow=n+1, ncol=p)
last_contacted[1,] <- sample(states, p)
colnames(last_contacted) <- states

# Compute the distance matrix for each state and potential state pairs
dist.m <- t(sapply(states, function(st)
  sapply(states, function(pot.st)
    dist_matrix[pot.st, last_contacted[1,st]])))
distance <- matrix(spline_function(as.vector(dist.m),
                                   fit),
                   nrow=p, ncol=p)

#simulation of events
i<-1
while (i <=n){

  if(i%%100==0) print(i)
  current.weekday<-1*(current.tm%%7>=2)
  hazard<- exp(b0*current.weekday+b1*rec+distance)/(p^2)

  # time increase
  dt<-rexp(1,sum(hazard))

  # check if this change meant that the event
  # moved from weekend to weekday or vice-versa:
  new.weekday<-1*((current.tm+dt)%%7>=2)
  if (dt>5| new.weekday!=current.weekday){
    current.tm<- (current.tm%/%7)*7+2+current.weekday*5
  } else {
    current.tm<-current.tm+dt
    tms<-c(tms,current.tm)
    event.id<-sample(1:p^2, 1,prob = hazard/sum(hazard))
    s<-(event.id-1)%%p+1
    r<-(event.id-1)%/%p+1
    dat<-rbind(dat,c(current.tm,s,r,event.id,
                     current.weekday,
                     dist.m[s,r],
                     rec[s,r]))
```

```
      non.event.id <- sample(setdiff(1:p^2, event.id),1)
      non.s <- (non.event.id-1)%%p+1
      non.r <- (non.event.id-1)%/%p+1
      nondat <- rbind(nondat,c(non.s,non.r,non.event.id,
                               dist.m[non.s,non.r],
                               rec[non.s,non.r]))

      rec[r,s] <- rec[r,s] + 1
      last_contacted[i+1,] <- last_contacted[i,]
      last_contacted[i+1,states[s]] <- states[r]

      dist.m <- t(sapply(states, function(st)
        sapply(states, function(pot.st)
          dist_matrix[pot.st, last_contacted[i+1,st]])))
      distance <- matrix(spline_function(as.vector(dist.m),
                                         fit),
                         nrow=p, ncol=p)
      hazard<- exp(b0*current.weekday+b1*rec+distance)/(p^2)

      i<-i+1
    }
  }

  remdat <- cbind(1:n, dat, nondat,
                  dat[,6]-nondat[,4],
                  dat[,7]-nondat[,5],
                  1)
  colnames(remdat) <- c("index", "tms","s","r","event.id",
                        "weekday","distance", "rec",
                        "non.s","non.r","non.event.id",
                        "non.distance", "non.rec",
                        "delta.dist", "delta.rec","one")
  remdat <- as.data.frame(remdat)

  return(list(remdat, last_contacted))
}
```

```
n_sim = 100
set.seed(1234)
remdats <- sapply(1:n_sim, function(x)
  event_simulator(states = states, p = length(states), n = n,
                  b0 = b0, b1 = b1, fit, seed=x),
  simplify =FALSE)
save(remdats, file="output/remdats_data.RData")
```

Model Fitting (no shifting)

For each of the simulated dataset, we fit a model including a linear effect fir reciprocity and a non-linear effect for distance. Particularly, for each event we did sample one non-event and the distance between the covariates is taken into account.

```
load(file="output/remdats_data.RData")
remdats_data <- lapply(remdats, function(x) x[[1]])
gam.fits <- lapply(remdats_data, function(x){
```

```
  dist_MAT <- cbind(x$distance, x$non.distance)
  ONE <- cbind(x$one, -x$one)
  fit <- gam(one ~ s(dist_MAT, by=ONE) +
               delta.rec - 1,
             family="binomial",
             data=x)
  return(fit)
}
)
save(gam.fits, file="output/gam.fits.RData")
```

Event Shifting

```
remdat_shifting <- function(remdat, last_contacted, p, n, seed){

  set.seed(seed)

  T<-max(remdat$tms)
  shifts<-rexp(p^2,1/T)
  remdat_complete<-NULL

  for (i in 1:n){

    current.tm <- remdat[i,2]

    atrisk<-(shifts<current.tm+shifts[remdat[i,5]]) &
            (shifts+T>current.tm+shifts[remdat[i,5]])

    non.event.id <- sample(1:p^2,1,prob = atrisk/sum(atrisk))
    non.s<-(non.event.id-1)%%p+1
    non.r<-(non.event.id-1)%/%p+1

    non.current.weekday<-1*((current.tm+shifts[remdat[i,5]]-
                              shifts[non.event.id])%%7>=2)
    non.rec<-sum((remdat[,2]<current.tm+shifts[remdat[i,5]]-
                    shifts[non.event.id])&(remdat[,3]==non.r)&
                   (remdat[,4]==non.s))

    dist.m <- t(sapply(states, function(st)
      sapply(states, function(pot.st)
        dist_matrix[pot.st, last_contacted[i+1,st]])))

    remdat_complete<-rbind(remdat_complete,
                           unlist(c(remdat[i,1:8],
                                    non.s,non.r,non.event.id,
                                    non.current.weekday,
                                    dist.m[non.s,non.r],non.rec)))
  }

  colnames(remdat_complete) <- c(colnames(remdat)[1:8],
                                 "non.s","non.r","non.event.id",
                                 "non.weekday","non.distance","non.rec")
  remdat_complete <- as.data.frame(remdat_complete)
```

```r
  remdat_complete$delta.dist = remdat_complete$distance -
    remdat_complete$non.distance
  remdat_complete$delta.wkday = remdat_complete$weekday -
    remdat_complete$non.weekday
  remdat_complete$delta.rec = remdat_complete$rec -
    remdat_complete$non.rec
  remdat_complete$one = 1

  return(remdat_complete)

}
```

```r
shifted_rems <- sapply(1:n_sim, function(x)
  remdat_shifting(remdats[[x]][[1]], remdats[[x]][[2]], p, n, seed=x),
  simplify = FALSE)
save(shifted_rems, file="output/shifted_rems.RData")
length(shifted_rems)
```

Model Fitting (shifting)

```r
load(file="output/shifted_rems.RData")
gam.fits_shifted <- lapply(shifted_rems, function(x){
  dist_MAT <- cbind(as.numeric(x$distance),
                    as.numeric(x$non.distance))
  ONE <- cbind(x$one, -x$one)
  fit <- gam(one ~ s(dist_MAT, by=ONE, k=11, bs="bs") +
               delta.rec + delta.wkday - 1,
             family="binomial",
             data=x)
  return(fit)
}
)
save(gam.fits_shifted, file="output/gam.fits_shifted.RData")
```